Experiment – 8:
Laplace Transform, Inverse Transform, ROC and Stability

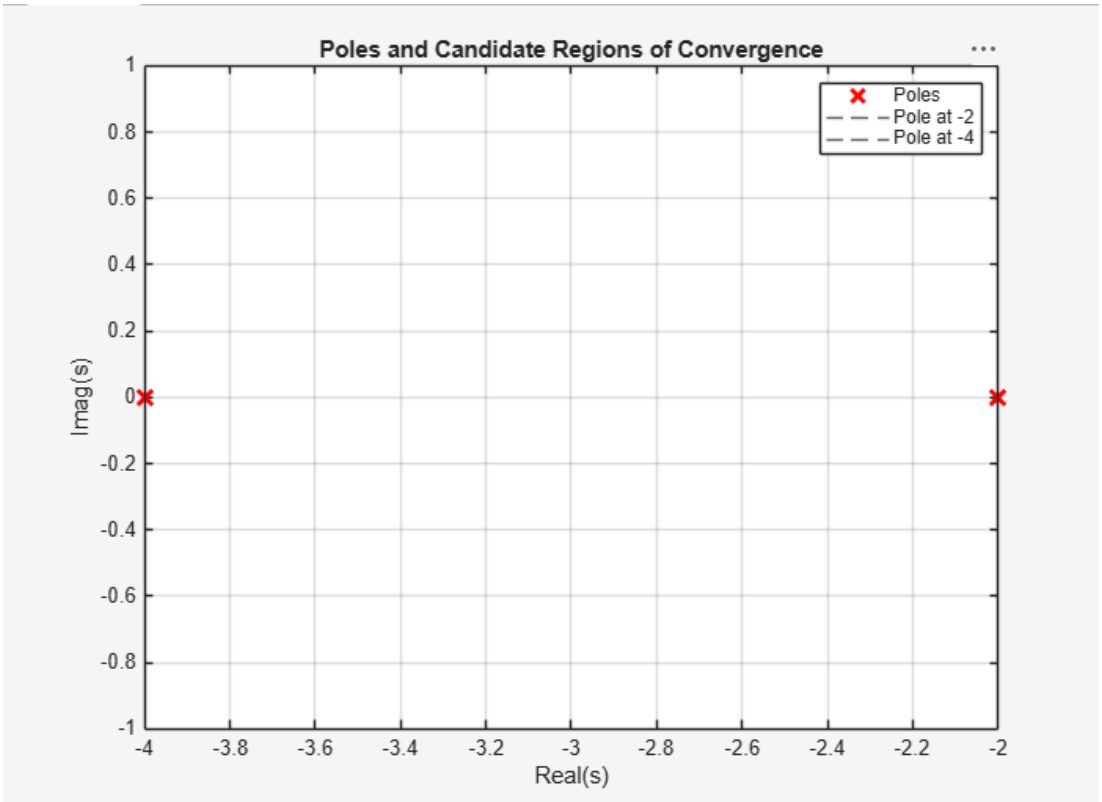
iii)

/MATLAB Drive/Experiment_8_3.m

```
clc; clear; close all;
%% 8(iii ) Analysis of Region of Convergence (ROC) and Its Significance in System

figure;
% ROC Analysis for a Rational Laplace Transform
% Requires Symbolic Math Toolbox
% X(s) = 1/((s+2)(s+4))
% Poles are at s = -2 and s = -4.
p = [-2 -4];
plot(real(p),imag(p),'rx','MarkerSize',10,'LineWidth',2);
hold on;
xline(-2,'--'); xline(-4,'--');
grid on;
xlabel('Real(s)'); ylabel('Imag(s)');
title('Poles and Candidate Regions of Convergence');
legend('Poles','Pole at -2','Pole at -4');
fprintf('Poles are at s = -2 and s = -4.\n');
fprintf('For a right-sided signal, ROC: Re(s) > -2.\n');
fprintf('For a left-sided signal, ROC: Re(s) < -4.\n');
fprintf('For a two-sided signal, ROC: -4 < Re(s) < -2.\n');
```

Code



Output